



UVAN SHANKAR S J

Full-Stack Engineer | Software Engineer

Software Engineer with 2 years of full-stack development and automation experience. Passionate about building secure, scalable web applications using React and Node.js. Combines a developer's mindset with rigorous testing expertise—including unit, integration, and CI/CD pipeline management—to deliver highly optimized, resilient, and production-ready code.

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SKILLS & LANGUAGES:

Languages: JavaScript (ES6+), Python, Ruby, C++, SQL, HTML5, CSS3.

Frontend: React.js, Redux, Context API, Tailwind CSS, Responsive Design.

Backend & Databases: Node.js, Express.js, RESTful API Design, JWT, MongoDB, MySQL (Sequelize), DynamoDB.

DevOps & Cloud: AWS (EC2, S3, Lambda), Docker, Jenkins (CI/CD), Git, Perforce, SonarQube.

Testing/Automation: Jest, Selenium, Computer Vision Tools.

Operating Systems: Linux (Ubuntu)

PROJECTS:

Expense Tracker Analytics App - (2025)([Link](#))

Tech Stack: React, Node.js, Express, MongoDB, REST APIs, Context API, Chart.js / Recharts

- Architected and deployed a full-stack financial analytics application featuring dynamic fund-allocation tracking, utilizing **React** for the user interface and **Node.js/Express** with **MongoDB** for the backend database."
- Advanced State Management & Component Design:**
"Leveraged the **React Context API** to manage global authentication and real-time expense data states, eliminating prop-drilling and ensuring clean, modular component separation."
- Performance Optimization & Code Splitting:**
"Optimized application bundle sizes and initial page-load velocity by implementing **React Lazy Loading** and code-splitting strategies for asynchronous dashboard rendering."
- Backend API & Data Aggregation:**
"Engineered a secure **RESTful API** layer to handle transaction CRUD operations, utilizing complex MongoDB aggregation queries to group, filter, and serve structured data to frontend graphing components."

Faculty management system— (2023) ([link](#))

- Built a MERN stack (MySQL, Express.js, React.js, Node.js) dashboard for workload tracking.
- Increased faculty productivity by 40% through streamlined process automation.
- Used Sequelize ORM and session management with JWT for secure data handling.
- Integrated API testing, performance testing, and front-end unit tests with tools like Selenium, Jest, and SonarQube.

Autonomous delivery robot— (2023) ([link](#))

- Created a web interface for sending/receiving robot navigation data via REST API.
- Integrated AWS cloud database services DynamoDB into project architecture to ensure scalable and secure data storage.
- Developed an instruction-based path delivery system; future work includes optimizing routes using ride data.

WORK EXPERIENCE:

Software Development Engineer in Test 1

ICU Medical ,Chennai |Jan 2024 - Present:

- Architected and led the development of a **Computer Vision-based automation platform** to interface with physical medical device interfaces, optimizing real-time device telemetry capturing and feedback processing.
- Engineered a modular **Python hardware-abstraction library** to control proprietary medical pump mechanics, streamlining cross-device data synchronization and cross-functional operational workflows
- Developed **bat** script to automate environment configuration and dependency management on **windows-based systems**, reducing developer onboarding time by 30% .
- Developed a Batch scripting utility to automate system state capture and diagnostic logging, streamlining data collection workflows and reducing manual entry errors.
- Designed and deployed a **Python data-migration utility leveraging the TestRail REST API** to programmatically batch-edit, sync, and transfer large-scale datasets across distinct enterprise repositories, eliminating manual configuration errors. .
- Managed complex codebases and hardware-software integration scripts using **Perforce**, ensuring version integrity across multi-functional engineering teams
- Collaborated in an Agile/Scrum framework to execute concurrent feature implementation, managing continuous integration (CI/CD) pipelines to rapidly validate and deploy critical updates

EDUCATION:

B.Tech in Computer Science and Engineering
Amrita Vishwa Vidyapeetham, Coimbatore – 2020 to 2024
CGPA: 7.31/10

Certification:

AWS Cloud Computing (Introduction to Cloud): ([link](#))

Learned and implemented the AWS cloud services like S3, EC2, Lambda, DynamoDB, CloudTrail, CloudWatch, IAM, CloudFront, and ElastiCache.

Additional:

Solved 130+ programming challenges on LeetCode using Python ([link](#))

Strong foundation in Data Structures, Algorithms, and Backend System Design.